

For questions 1-2, draw a representation of the expression with algebra tiles. Cross out any zero pairs. Then write an equivalent algebraic expression after eliminating any zero pairs.

	Algebraic Expression and Representation	Equivalent Algebraic Expression
1.	$-8x - 2 + 6x$ Representation: Diagram showing algebra tiles for $-8x - 2 + 6x$. The top row consists of 8 negative x tiles ($-x$). The middle row consists of 6 positive x tiles ($+x$). The bottom row consists of 2 negative unit tiles (-1). Red lines connect 6 negative x tiles to 6 positive x tiles, forming zero pairs. The remaining 2 negative x tiles and 2 negative unit tiles are shown below.	$-2x - 2$
2.	$10 + z - 6 + z$ Representation: Diagram showing algebra tiles for $10 + z - 6 + z$. The top row consists of 10 positive unit tiles ($+1$). The middle row consists of 6 negative unit tiles (-1). The bottom row consists of 2 positive z tiles ($+z$). Red lines connect 6 positive unit tiles to 6 negative unit tiles, forming zero pairs. The remaining 4 positive unit tiles and 2 positive z tiles are shown below.	$2z + 4$

Write an equivalent expression by combining like terms for questions 3-10.

3. $4x + 9x$

$13x$

4. $-2y + 5 + y - 7$

$-2y + y - 7 + 5$

$-y - 2$

5. $r + -r + r - r$

$r + r - r - r$

0

6. $6y - 3 + 8 - 4y$

$6y - 4y + 8 - 3$

$2y + 5$

7. $y - 4 + 5y + 8$

$y + 5y + 8 - 4$

$6y + 4$

8. $10x + 9 + x - 1$

$10x + x + 9 - 1$

$11x + 8$

9. $-5y + 2 + 3 + 9y$

$9y - 5y + 2 + 3$

$4y + 5$

10. $8 + z - 10 + 2z$

$z + 2z - 10 + 8$

$3z - 2$